

1.使用general log 进行审计

MySQL Workbench

Local instance MySQL80 x unconnected x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

customer_info

Tables

Views

Stored Procedures

Functions

mysql

Tables

course

sc

student

Views

Stored Procedures

Functions

sys

work2

world

Administration Schemas

Information

Schema: mysql

Object Info Session

Query Completed

SQL File 3 course student sc

Limit to 2000 rows

172

173

174

175 • show global variables like 'log_general';

176 • set global general_log = 'ON';

177 • show global variables like 'log_output';

178 • set global log_output = 'TABLE';

179

180 • select * from sc;

181

182 • drop table sc;

183

184 • select * from mysql.general_log;

185 • select event_time,user_host,cast(argument as char)as decoded_arg from mysql.general_log;

186

187

188

189

190

191

Result Grid

Filter Rows

Exports

Wrap Cell Contents

event_time	user_host	thread_id	server_id	command_type	argument
2023-11-21 16:29:16.413925	root[root] @ localhost [::1]	9	1	Query	show global variables like 'log_general';
2023-11-21 16:29:16.414273	root[root] @ localhost [::1]	9	1	Query	set global general_log = 'ON';
2023-11-21 16:29:16.415194	root[root] @ localhost [::1]	9	1	Query	show global variables like 'log_output';
2023-11-21 16:31:19.881361	root[root] @ localhost [::1]	10	1	Query	set global log_output = 'TABLE';
2023-11-21 16:31:19.888777	root[root] @ localhost [::1]	9	1	Query	select * from sc;
2023-11-21 16:31:19.889644	root[root] @ localhost [::1]	9	1	Query	drop table sc;
2023-11-21 16:31:19.890171	root[root] @ localhost [::1]	9	1	Query	select * from mysql.general_log;
2023-11-21 16:32:43.560232	root[root] @ localhost [::1]	10	1	Query	select event_time,user_host,cast(argument as char)as decoded_arg from mysql.general_log;

general_log 7 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
45	16:31:19	select event_time,user_host,cast(argument as char)as decoded_arg from mysql.general_log LIMIT 0, 2000	116 row(s) returned	0.000 sec / 0.000 sec
46	16:32:43	select * from mysql.general_log LIMIT 0, 2000	120 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80 x unconnected x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

customer_info

Tables

Views

Stored Procedures

Functions

mysql

Tables

course

sc

student

Views

Stored Procedures

Functions

sys

work2

world

Administration Schemas

Information

Schema: mysql

Object Info Session

Query Completed

SQL File 3 course student sc

Limit to 2000 rows

172

173

174

175 • show global variables like 'log_general';

176 • set global general_log = 'ON';

177 • show global variables like 'log_output';

178 • set global log_output = 'TABLE';

179

180 • select * from sc;

181

182 • drop table sc;

183

184 • select * from mysql.general_log;

185 • select event_time,user_host,cast(argument as char)as decoded_arg from mysql.general_log;

186

187

188

189

190

191

Result Grid

Filter Rows

Exports

Wrap Cell Contents

event_time	user_host	decoded_arg
2023-11-21 16:28:42.799036	root[root] @ localhost [::1]	SELECT st.* FROM performance_schema.event...
2023-11-21 16:28:42.799970	root[root] @ localhost [::1]	SHOW INDEX FROM 'mysql' . 'sc' ;
2023-11-21 16:29:16.405195	root[root] @ localhost [::1]	select * from mysql.general_log LIMIT 0, 2000
2023-11-21 16:29:16.413179	root[root] @ localhost [::1]	SELECT st.* FROM performance_schema.event...
2023-11-21 16:29:16.413925	root[root] @ localhost [::1]	SELECT st.* FROM performance_schema.event...
2023-11-21 16:29:16.414273	root[root] @ localhost [::1]	SELECT st.* FROM performance_schema.event...
2023-11-21 16:29:16.415194	root[root] @ localhost [::1]	SHOW INDEX FROM 'mysql' . 'general_log' ;
2023-11-21 16:31:19.881361	root[root] @ localhost [::1]	select event_time,user_host,cast(argument as ...

Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
44	16:29:16	select * from mysql.general_log LIMIT 0, 2000	111 row(s) returned	0.000 sec / 0.000 sec
45	16:31:19	select event_time,user_host,cast(argument as char)as decoded_arg from mysql.general_log LIMIT 0, 2000	116 row(s) returned	0.000 sec / 0.000 sec

2.使用binlog+init_connect进行审计

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL statements:

```
1  
2  
3 • create user U1@localhost identified by '123456';  
4 • create user U2@localhost identified by 'abcdefg';  
5  
6 • grant all on jwgl to U1@localhost;  
7 • grant all on jwgl to U2@localhost;  
8  
9  
10 • show grants for U1@localhost ;  
11 • show grants for U2@localhost ;  
12  
13  
14  
15 • create database conn_db;  
16 • create table conn_db.logintable(  
17   id int not null auto_increment,  
18   ... )
```

The 'Result Grid' shows the output of the 'show grants for U1@localhost' query:

Grants for U1@localhost			
▶	GRANT USAGE ON *.* TO 'U1'@'localhost'		
	GRANT INSERT ON 'conn_db'. 'logintable' TO ...		
	GRANT ALL PRIVILEGES ON 'jwgl'. 'jwgl' TO ...		
	GRANT ALL PRIVILEGES ON 'work2'. 'jwgl' TO...		

The 'Output' window shows the 'Action Output' table:

#	Time	Action	Message	Duration / Fetch
✓ 13	11:04:26	grant all on jwgl to U2@localhost	0 row(s) affected	0.016 sec
✓ 14	11:05:32	show grants for U1@localhost	4 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- customer_info
- jwtl
 - Tables
 - course
 - sc
 - student
 - Views
 - Stored Procedures
 - Functions
- sys
- work2
- world

Administration Schemas

Information

No object selected

Query 1 x course sc student

Limit to 2000 rows

```
1
2
3 • create user U1@localhost identified by '123456';
4 • create user U2@localhost identified by 'abcdefg';
5
6 • grant all on jwtl to U1@localhost;
7 • grant all on jwtl to U2@localhost;
8
9
10 • show grants for U1@localhost ;
11 • show grants for U2@localhost ;
12
13
14
15 • create database conn_db;
16 • create table conn_db.logintable(
17   id int not null auto_increment,
```

Result Grid

Filter Rows:

Exports

Wrap Cell Content:

Result Grid

Form Editor

Result 4 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 14	11:05:32	show grants for U1@localhost	4 row(s) returned	0.000 sec / 0.000 sec
✓ 15	11:06:06	show grants for U2@localhost	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

conn_db

Tables

logintable

Views

Stored Procedures

Functions

customer_info

jwgl

sys

work2

world

Administration Schemas

Information

Table: logintable

Columns:

id int AI PK

thread_id int

login_time timestamp

localname varchar(50)

Query 1 x course sc student logintable

Limit to 2000 rows

```
8
9
10 • show grants for U1@localhost ;
11 • show grants for U2@localhost ;
12
13
14
15 • create database conn_db;
16 • create table conn_db.logintable(
17     id int not null auto_increment,
18     thread_id int not null,
19     login_time timestamp,
20     localname varchar(50) default null,
21     primary key(id)
22 );
23 • SELECT * FROM conn_db.logintable;
24
```

Result Grid

Filter Rows:

Edit

Export/Import

Wrap Cell Content

	id	thread_id	login_time	localname
*	NULL	NULL	NULL	NULL

logintable 6 x

Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 18	11:09:44	SELECT * FROM conn_db.logintable LIMIT 0...	0 row(s) returned	0.000 sec / 0.000 sec
✓ 19	11:09:58	SELECT * FROM conn_db.logintable LIMIT 0...	0 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- conn_db
 - Tables
 - logintable
 - Views
 - Stored Procedures
 - Functions
- customer_info
- jwgl
- sys
- work2
- world

Administration Schemas

Information

Table: **logintable**

Columns:

id	int AI PK
thread_id	int
login_time	timestamp
localname	varchar(50)

Query 1 x course sc student logintable

Limit to 2000 rows

```

9
10 • show grants for U1@localhost ;
11 • show grants for U2@localhost ;
12
13
14
15 • create database conn_db;
16 • create table conn_db.logintable(
17   id int not null auto_increment,
18   thread_id int not null,
19   login_time timestamp,
20   localname varchar(50) default null,
21   primary key(id)
22 );
23 • SELECT * FROM conn_db.logintable;
24
25 • grant insert on conn_db.logintable to U1@localhost;
26 • grant insert on conn_db.logintable to U2@localhost;
27
28 • show variables like 'init_connect%';
29 • set global init_connect='insert into conn_db.logintable(thread_id,log_time,lo
30   values(connection_id(),now(),user(),current_user())';
31
32 • show global variables like 'log_bin';
33
34
  
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 20	11:10:22	grant insert on conn_db.logintable to U1@loc...	0 row(s) affected	0.016 sec
✓ 21	11:10:24	grant insert on conn_db.logintable to U2@loc...	0 row(s) affected	0.000 sec

Object Info Session

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

conn_db

Tables

loginable

Views

Stored Procedures

Functions

customer_info

juvj

sys

work2

world

Administration Schemas Information

Table: loginable

Columns:

id int AI PK

thread_id int

login_time timestamp

localname varchar(50)

Query 1 course sc student loginable

Limit to 2000 rows

```
8
9
10 • show grants for U1@localhost ;
11 • show grants for U2@localhost ;
12
13
14
15 • create database conn_db;
16 • create table conn_db.loginable(
17   id int not null auto_increment,
18   thread_id int not null,
19   login_time timestamp,
20   localname varchar(50) default null,
21   primary key(id)
22 );
23 • SELECT * FROM conn_db.loginable;
24
25 • grant insert on conn_db.loginable to U1@localhost;
26 • grant insert on conn_db.loginable to U2@localhost;
27
28 • show variables like 'init_connect';
29 • set global init_connect='insert into conn_db.loginable(thread_id,log_time,localname,matchname)
30   values(connection_id(),now(),user(),current_user())';
31
32 • show global variables like 'log_bin';
33
34
```

Output

#	Time	Action	Message	Duration / Fetch
21	11:10:24	grant insert on conn_db.loginable to U2@localhost	0 row(s) affected	0.000 sec
22	11:10:42	set global init_connect='insert into conn_db.loginable(thread_id,log_time,localname,matchname) values(connection_id(),now(),...'	0 row(s) affected	0.000 sec

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- customer_info
- juwl
 - Tables
 - course
 - sc
 - student
 - Views
 - Stored Procedures
 - Functions
- sys
- work2
- world

Administration Schemas

Information

No object selected

Query 1 x course sc student

Limit to 2000 rows

```
16 • create table conn_db.logintable(  
17     id int not null auto_increment,  
18     thread_id int not null,  
19     login_time timestamp,  
20     localname varchar(50) default null,  
21     primary key(id)  
22 );  
23  
24 • grant insert on conn_db.logintable to U1@localhost;  
25 • grant insert on conn_db.logintable to U2@localhost;  
26  
27 • show variables like 'init_connect';  
28 • set global init_connect='insert into';  
29  
30 • show global variables like 'log_bin';  
31  
32
```

Result Grid

Variable_name	Value
log_bin	ON

Result 5 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 15	11:06:06	show grants for U2@localhost	4 row(s) returned	0.000 sec / 0.000 sec
✓ 16	11:06:34	show global variables like 'log_bin'	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

管理员: C:\WINDOWS\system32\cmd.exe - mysql -u U1 -p

C:\Windows\System32>net start mysql80

请求的服务已经启动。

请键入 NET HELPMSG 2182 以获得更多的帮助。

C:\Windows\System32>mysql -u u1 -p

Enter password: *****

ERROR 1045 (28000): Access denied for user 'u1'@'localhost' (using password: YES)

C:\Windows\System32>mysql -u U1 -p

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 14

Server version: 8.0.35 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use jwgl;

Database changed

mysql> insert into course(cno,cname) values('202009081','高级数据库技术');

Query OK, 1 row affected (0.01 sec)

mysql>

3.MySql数据加解密

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

conn_db

- Tables
 - logintable
- Views
- Stored Procedures
- Functions

customer_info

jwgl

sys

work2

world

- Tables
- Views
- Stored Procedures
- Functions

Administration

Schemas

Information

Table: logintable

Columns:

- id int AI PK
- thread_id int
- login_time timestamp
- localname varchar(50)

Query 1 x

course sc student logintable

Limit to 2000 rows

36

37 • USE world;

38

39 • create table LoginInfo(

- 40 id int not null auto_increment,
- 41 userName varchar(10),
- 42 password blob,
- 43 primary key(id)
- 44);

45

46 • insert into LoginInfo values('01','U1',md5('123456'));

47 • insert into LoginInfo values('02','U2',SHA1('13579'));

48 • insert into LoginInfo values('03','U3',SHA2('02468',256));

49

50 • delete from LoginInfo where id = '03';

51 • SELECT * FROM LoginInfo;

52

53

54

55

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Result Grid

Form Editor

	id	userName	password
▶	1	U1	BLOB
	2	U2	BLOB
	3	U3	BLOB
•	NULL	NULL	NULL

LoginInfo 14 x

Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓	44 12:12:10	insert into LoginInfo values('03','U3',SHA2('02...	1 row(s) affected	0.000 sec
✓	45 12:12:11	SELECT * FROM LoginInfo LIMIT 0, 2000	3 row(s) returned	0.000 sec / 0.000 sec

Object Info

Session

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- conn_db
 - Tables
 - logintable
 - Views
 - Stored Procedures
 - Functions
- customer_info
- jwtl
- sys
- work2
- world
 - Tables
 - Views
 - Stored Procedures
 - Functions

Administration Schemas

Information

Table: **logintable**

Columns:

id	int AI PK
thread_id	int
login_time	timestamp
localname	varchar(50)

Query 1 x course sc student logintable

Limit to 2000 rows

```

41  userName varchar(10),
42  password blob,
43  primary key(id)
44  );
45
46  • insert into LoginInfo values('01','U1',md5('123456'));
47  • insert into LoginInfo values('02','U2',SHA1('13579'));
48  • insert into LoginInfo values('03','U3',SHA2('02468',256));
49
50  • delete from LoginInfo where id = '03';
51  • SELECT * FROM LoginInfo;
52
53  • insert into LoginInfo values('04','U4',aes_encrypt('987654','keystring'));
54
55  • select id,aes_decrypt(password,'keystring')as password
56  from LoginInfo where id = '04';
57
58
59
60

```

Result Grid

	id	password
▶ 4		BLOB

Result 15 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✗ 47	12:16:05	select id,aes_decrypt(pwd,keystring)as pass...	Error Code: 1054. Unknown column 'pwd' in fi...	0.000 sec
✓ 48	12:16:23	select id,aes_decrypt(password,keystring)as ...	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session